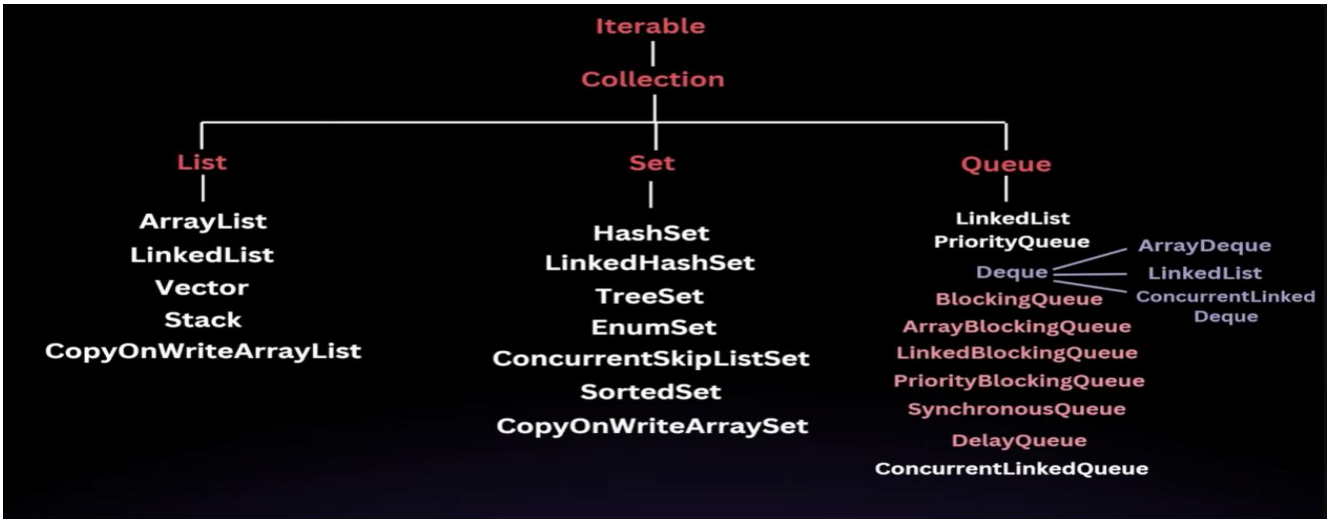


COLLECTIONS IN JAVA

Introduction to Java Collections Framework (Important Notes)

The **Java Collections Framework (JCF)** is a unified architecture used to **store, manipulate, and process groups of objects** efficiently. It provides ready-made data structures and algorithms, reducing the need to write complex data-handling logic from scratch. Collections are used extensively in **backend applications, Spring Boot, data processing, and APIs**.



🔴 **Map is not a child of Collection, but part of the framework.**

Collection Interface

Collection is the root interface that represents a group of objects. It defines basic operations such as add(), remove(), size(), and iterator() and is extended by List, Set, and Queue.

- HashMap – fastest, unordered
- LinkedHashMap – maintains insertion order
- TreeMap – sorted by key

🔴 Used when data must be **retrieved using a key** (e.g., id → employee).

List Interface

- ordered elements
- allows duplicates.

Common implementations:

- ArrayList – *fast access, most commonly used*
- LinkedList – *fast insertion/deletion*

Algorithms (Utility Class)

The Collections class provides algorithms such as:

- sort()
- reverse()
- shuffle()
- max() and min()

🔴 These methods operate on Collection objects.

Set Interface

- **unique elements only** (no duplicates).

Common implementations:

- HashSet – fastest, no order
- LinkedHashSet – maintains insertion order
- TreeSet – sorted order

🔴 Used when **uniqueness is required** (e.g., unique email IDs).

Where Collections are Used

- Storing database records
- Handling API responses
- Validation error lists
- Service-layer data processing
- Stream API operations

Queue Interface

- FIFO (First In First Out) principle.

Common implementations:

- PriorityQueue – priority-based processing
- ArrayDeque – double-ended queue

🔴 Used in task scheduling, messaging, and job processing.

Key Points to Remember

- Collections store **objects only**
- Prefer **interfaces over implementations**
- Choose collection type based on **order, uniqueness, and access speed**
- Collections work as the **foundation for Streams and Lambdas**

Map Interface

Map stores data in **key–value pairs**.

Common implementations:

One-Line Summary

The Java Collections Framework provides structured, reusable data structures like List, Set, Queue, and Map to efficiently store and process groups of objects in Java applications.

LIST

- This is SubInterface of Collection
- It Extends Collections
- Ordered
- Allows Duplicate
- Can Store Heterogeneous elements
- Implemented by – ArrayList , LinkedList , Stack , CopyOnWrite